

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T57 · Venn and Tree Diagrams

Questions

Student book · questions with writing space

■ 19 pages

FAMILY MAP

Probability Diagrams Venn Tree Diagrams

F01

Construct and read probability trees

7 Q | 18 marks

F02

Represent and read events with Venn diagrams

8 Q | 35 marks

Working Unit 10

4MA1-1H Q14

UNIT 1
4 marks

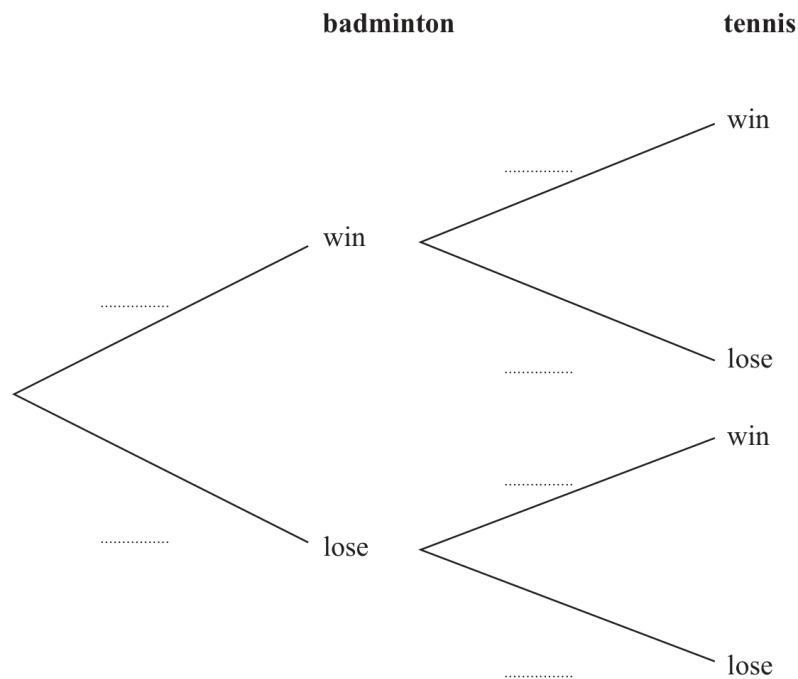
- 14 Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7

The probability that he will win the game of tennis is 0.4

- (a) Complete the probability tree diagram.



(2)

- (b) Work out the probability that Ricardo loses both games.

.....

(2)

Question 24

4MA1-2H Q16

MODULAR UNIT 1

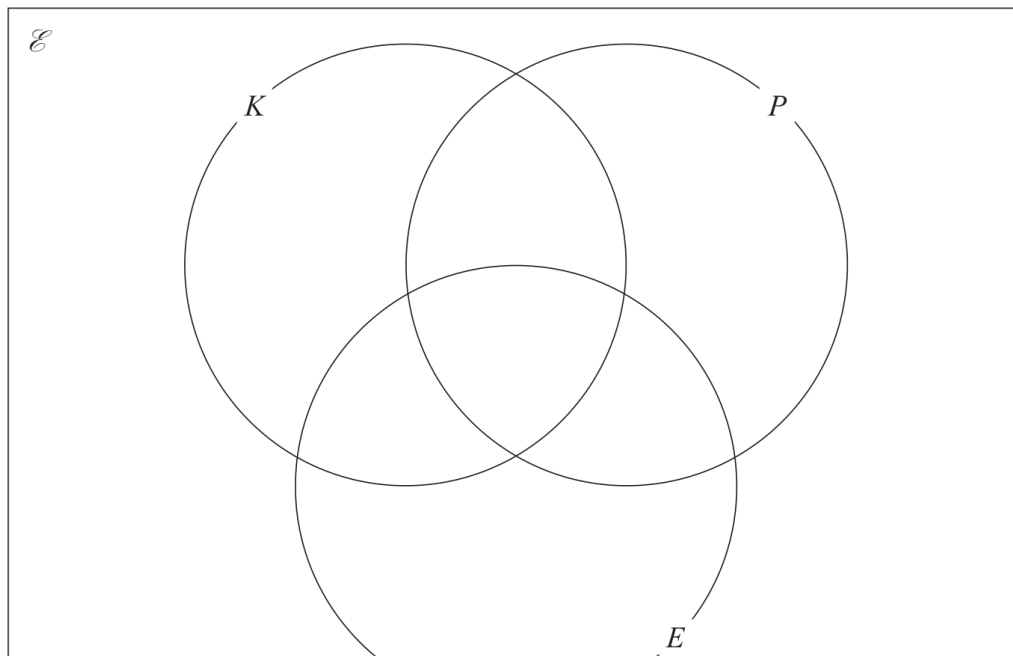
7 marks

- 16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

- (a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



Question 24 (continued)

4MA1-2H Q16

MODULAR UNIT 1

7 marks

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

.....
(2)

(c) Find $n(P \cap K')$

.....
(1)

(d) Find $n([P \cup E] \cap K)$

Question 10

4MA1-1H Q13 M01, M02

MODULAR UNIT 1

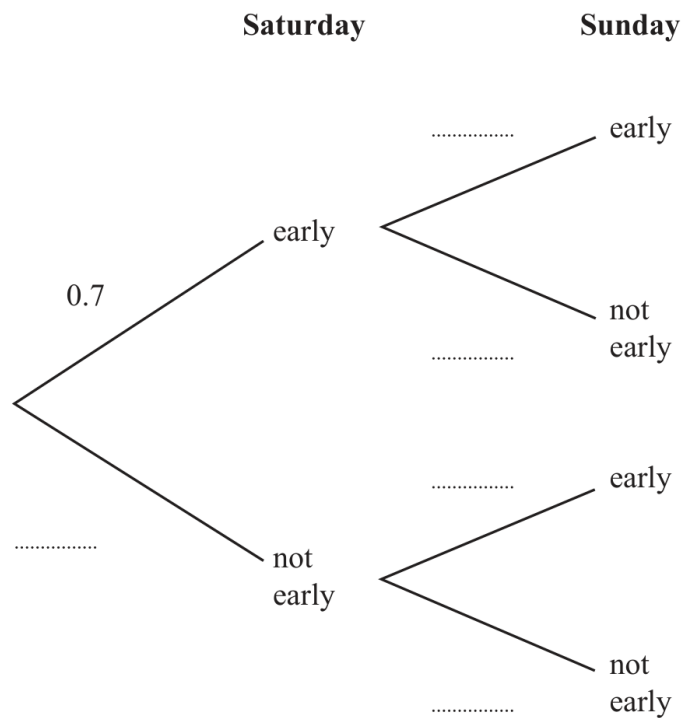
4 marks

- 13** The probability that Thomas gets to work early on Saturday is 0.7

If Thomas gets to work early on Saturday, the probability that he will get to work early on Sunday is 0.9

If Thomas does **not** get to work early on Saturday, the probability that he will get to work early on Sunday is 0.6

- (a) Use this information to complete the probability tree diagram.



(2)

- (b) Work out the probability that Thomas gets to work early on both Saturday and Sunday.

Question 12

4MA1-1H Q14 Q14(a)

MODULAR UNIT 1

2 marks

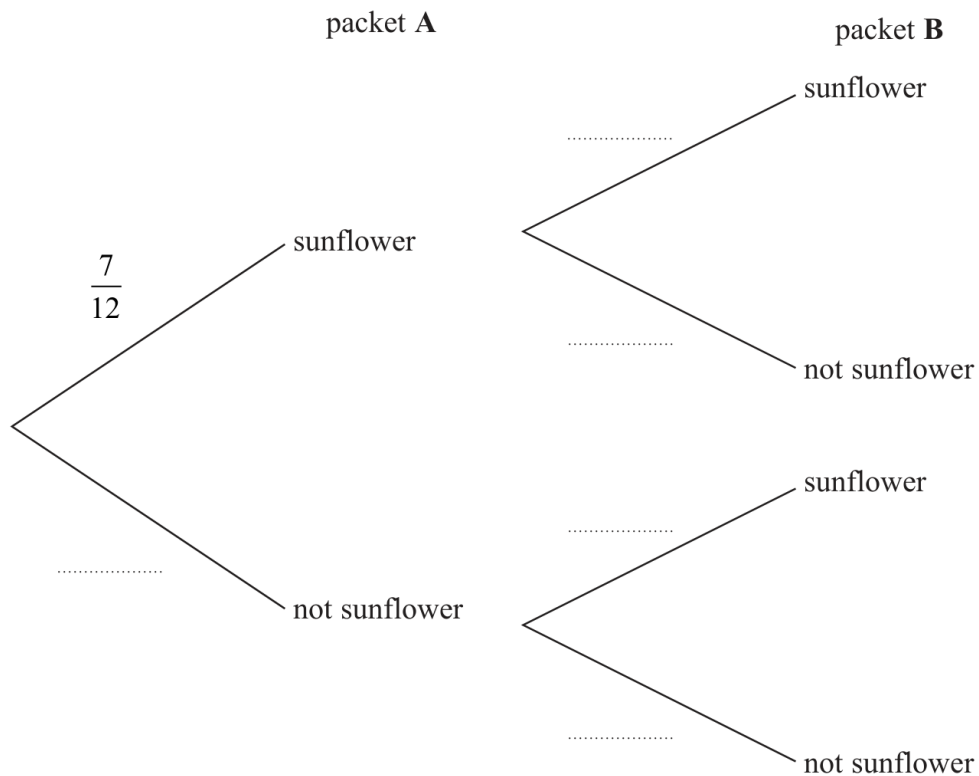
14 Aika has 2 packets of seeds, packet **A** and packet **B**

There are 12 seeds in packet **A** and 7 of these are sunflower seeds.

There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

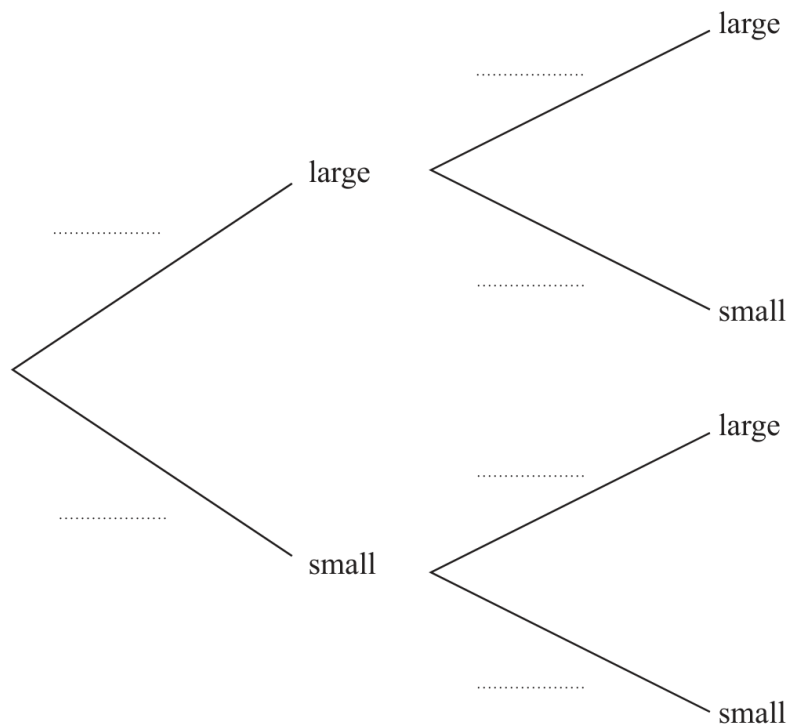
Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.



13 of the glasses are large
7 of the glasses are small

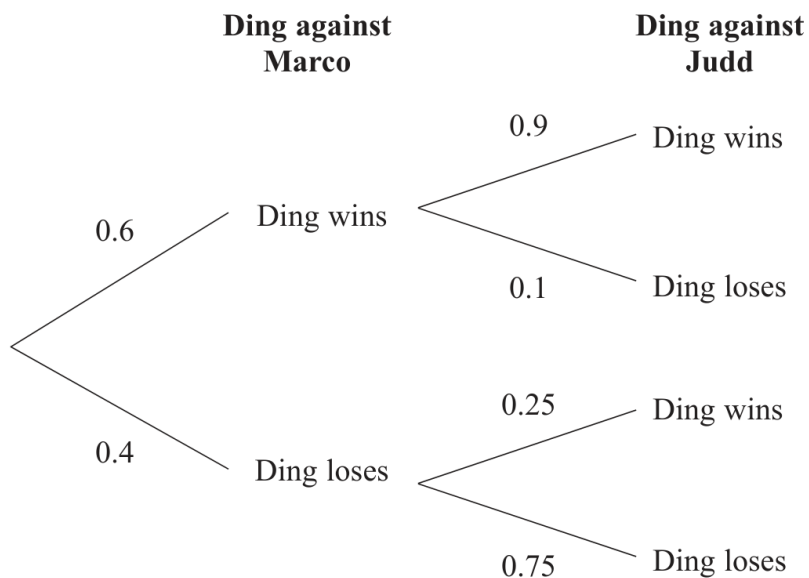
(a) Complete the probability tree diagram.



(b) Work out the probability that Roberto takes two small glasses.

- 14 Ding is going to play one game of snooker against each of two of his friends, Marco and Judd.

The probability tree diagram gives information about the probabilities that Ding will win or lose each of these two games.

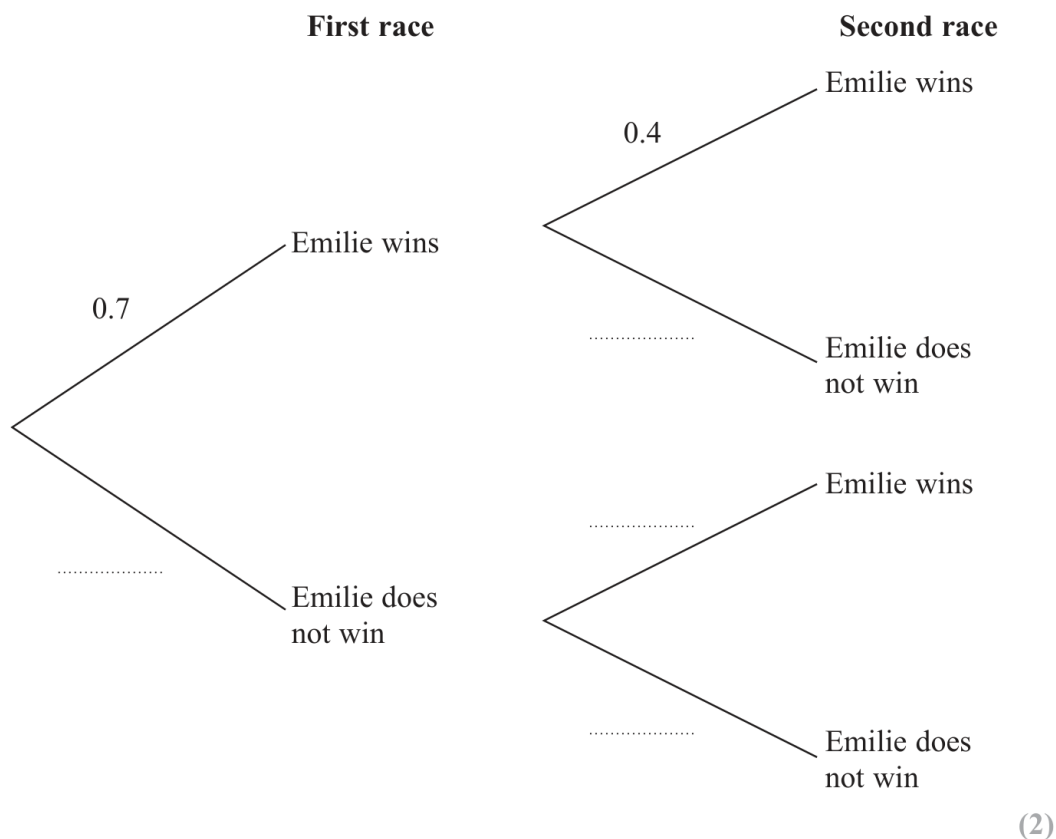


- (a) Work out the probability that Ding will win both games.

13 Emilie takes part in two races.

The probability that she wins the first race is 0.7
 The probability that she wins the second race is 0.4
 The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(b) Work out the probability that Emilie wins exactly one of the two races.

(3)

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

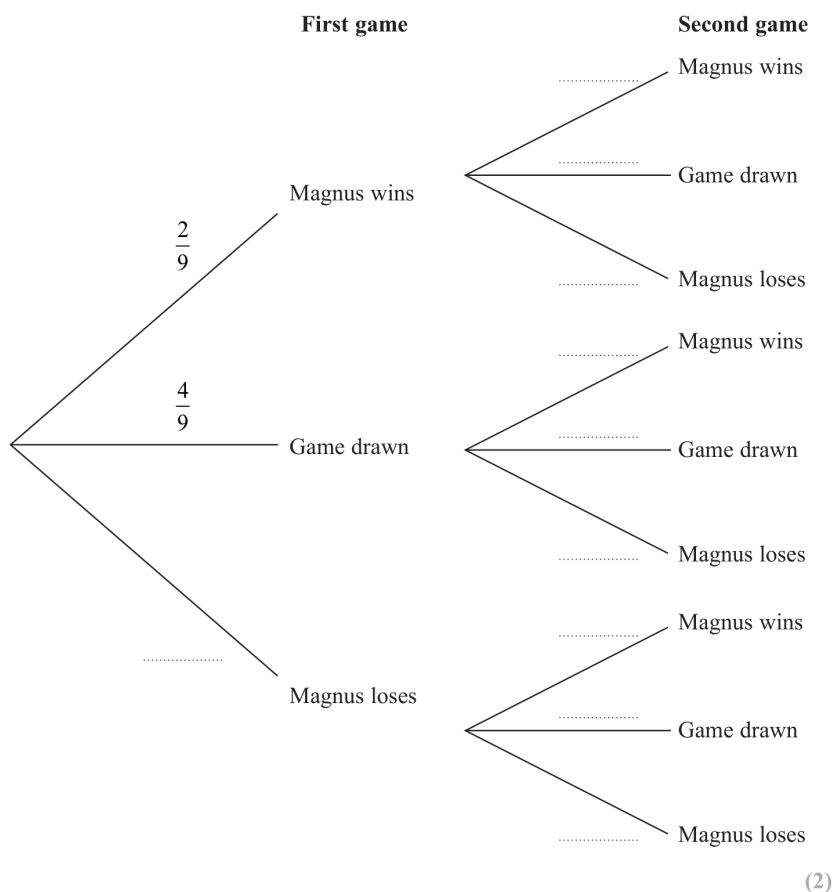
15 Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

(a) Complete the probability tree diagram.



For each game of chess,

the winner gets 2 points and the loser gets 0 points,
when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

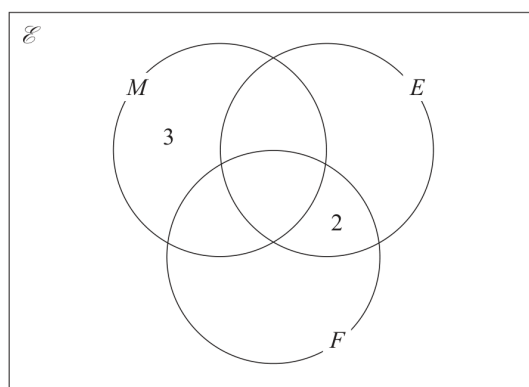
(3)

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

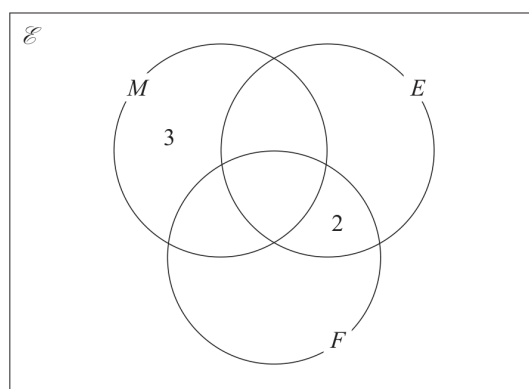
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



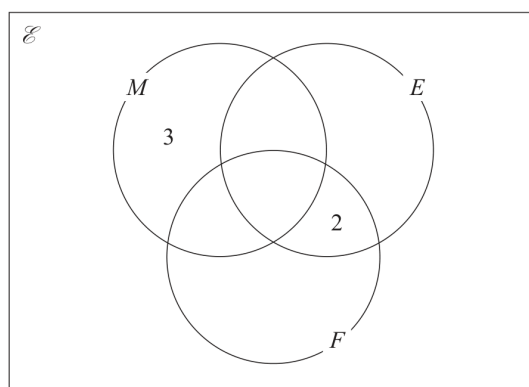
(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

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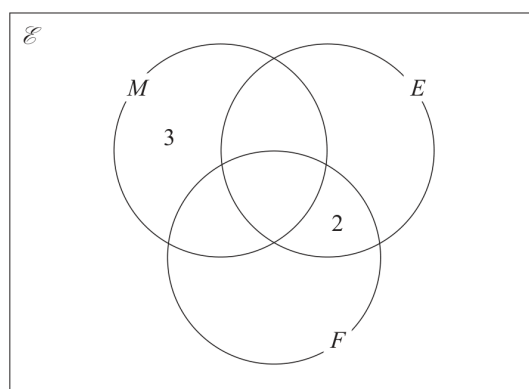
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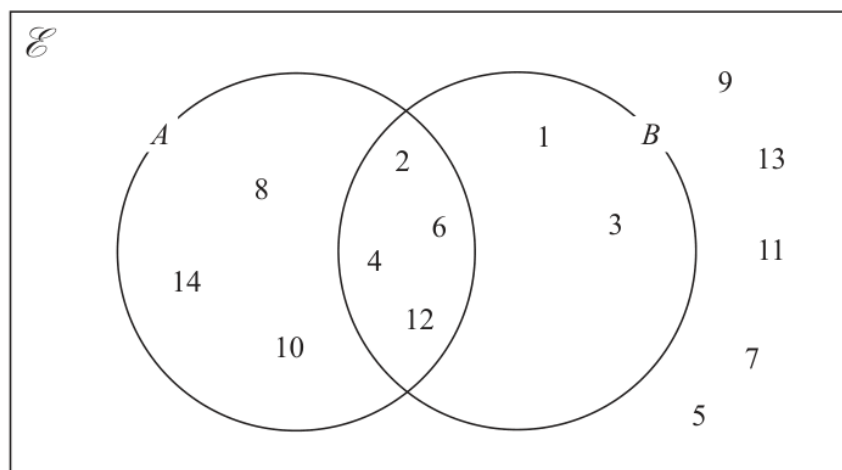


(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

- 1 The numbers from 1 to 14 are shown in the Venn diagram.



- (a) List the members of the set $A \cap B$

(1)

- (b) List the members of the set B'

(1)

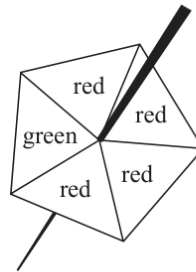
A number is picked at random from the numbers in the Venn diagram.

- (c) Find the probability that this number is in set A but is **not** in set B .

(2)

(Total for Question 1 is 4 marks)

Spinner A

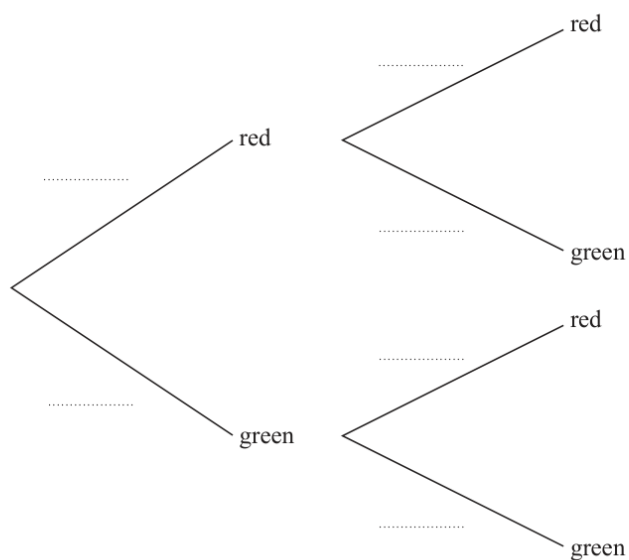


Spinner B

(a) Complete the probability tree diagram.

Spinner A

Spinner B



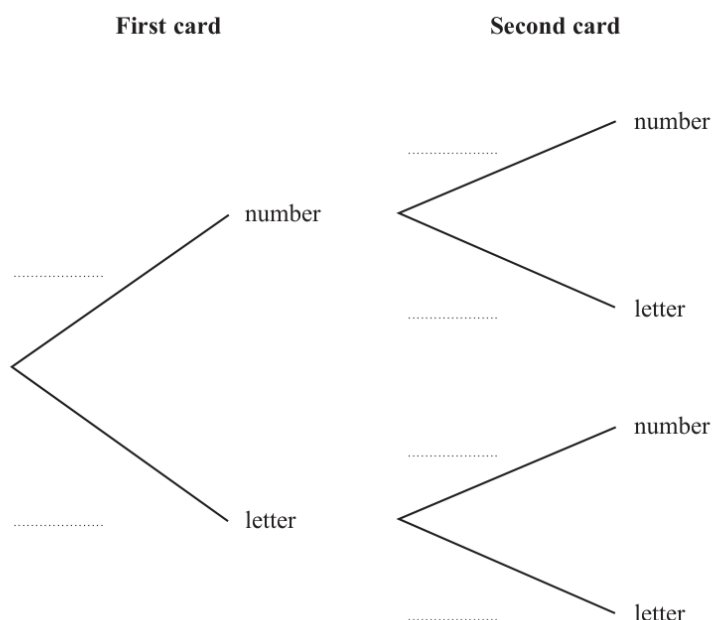
(b) Work out the probability that at least one of the spinners will land on green.

.....
(3)

(Total for Question 15 is 5 marks)

A	7	H	F	E	D	G	5	C
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(a) Complete the probability tree diagram.



(b) Work out the probability that each of the two cards has a number on it.

.....
(2)

(c) Work out the probability that there will be one card with a number on it and one card with a letter on it.

.....
(3)

.....
(Total for Question 15 is 7 marks)

Chi takes at random a bead from the bag and a bead from the box.

Bag

$\frac{2}{5}$

yellow

orange

Box

yellow

green

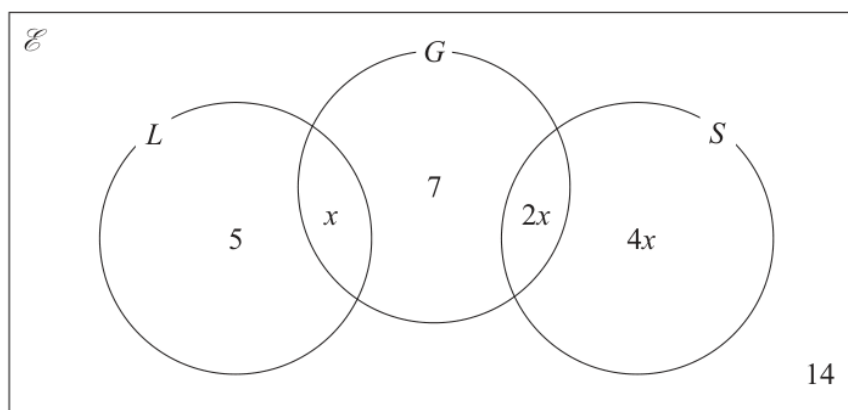
yellow

green

(b) Work out the probability that Chi takes two beads of different colours.

- 18** Bianca asked 40 students which of the languages Latin (L), Greek (G) and Sanskrit (S) they study.

The Venn diagram gives some information about her results.
It shows the number of students in each subset.



One of these students is selected at random.

Work out the probability that this student studies Latin or Sanskrit.